METRIC

MIL-DTL-32648 (USAF)
30 March 2020

DEPARTMENT OF DEFENSE
DETAIL SPECIFICATION

ZINC NICKEL PLATING, LOW HYDROGEN EMBRITTLEMENT ELECTRO-DEPOSITION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

AMSC: NA

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FOREWORD

1. This detail specification is approved for use by the AF-70 Hill AF Base, AFMC 417 SCMS/GUEA Landing Gear, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

2. This detail specification provides guidance for the repair process, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

3. Comments, suggestions, or questions on this specification should be addressed to the USAF Landing Gear Program MIL-DTL-32648 Content Manager, AFMC 417 SCMS/GUEA, Hill AFB, UT 84056-5228 or emailed to 417SCMS.GUEA.workflow@us.af.mil. Since contact information can change, verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This detail specification provides guidance on the procedures and processes for (LHE) Low Hydrogen Embrittling Zinc-nickel (Zn-Ni) plating applied by electro-deposition for the repair of existing parts or the procurement of new high strength steel substrates (180,000 psi or higher ultimate strength). Subsequent heat treating techniques needed to ensure low embrittlement of steel are also described. The process was developed specifically for high strength steel. However, it can also be used on lower strength steel alloys, corrosion resistant steel alloys, copper-based alloys, or other alloys as directed by technical data.

1.2 Classification. LHE Zn-Ni plating covered by this specification and specified in the technical data, are the following types and classes:

1.2.1 Types. LHE Zn-Ni plating is the following types:

Type I	As-plated without supplementary treatment.
Type II	As-plated with supplementary conversion coat treatment.
No Type	If no Type is specified, apply *Type II*.

1.2.2 Classes. LHE Zn-Ni plating is the following classes:

Class 1	0.013mm (0.0005 inch) thick minimum
Class 2	0.008mm (0.0003 inch) thick minimum
Class 3	0.005mm (0.0002 inch) thick minimum
No Class	If no class is specified in the technical data, apply *Class 2*.

NOTE: Unless otherwise specified, the maximum thickness for each class is 0.0076mm (0.0003 inches) greater than the minimum for the class.

1.3 Safety – Hazardous. While the materials, methods, applications, and processes described or referenced in this specification may involve the use of hazardous materials, this specification does ***not*** address the hazards, which may be involved in such use. It is the sole responsibility of the user to ensure familiarity with the safe and proper use of any hazardous materials and to take necessary precautionary measures to ensure familiarity with the safe and proper use of any hazardous materials and to take necessary precautionary measures to ensure the health and safety of all personnel involved.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited within this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. If the solicitation or contract does not specify an issue or revision, follow the current version of the applicable document(s).

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DEPARTMENT OF DEFENSE STANDARDS

MIL-DTL-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-DTL-85891	Engineered Media for Removal of Organic Coatings
MIL-PRF-23377	Primer Coatings: Epoxy, High-Solids
MIL-PRF-29602	Cleaning Compounds, Parts Washer, and Spray Cabinet
MIL-PRF-85582	Primer Coatings: Epoxy, Waterborne
MIL-STD-1504	Abrasive Blasting of Aircraft Components
MIL-STD-1916	DoD Preferred Methods for Acceptance of Product
MIL-STD-865	Selective, Brush Plating, Electro Deposition
MIL-STD-871	Electro-Chemical Stripping of Inorganic Finishes

(Copies of these documents are available online at https://assist.dla.mil.)

FEDERAL STANDARDS

FED-STD-313	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

29 CFR 1910.1200	Occupational Safety and Health Standards: Toxic and Hazardous Substances

(A copy of this document is available online at https://www.osha.gov/.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

AMS 2430	Shot Peening, Automatic
AMS 2451/9B	Plating, Brush, Tin-Zinc Low Hydrogen Embrittlement
AMS 2451	Plating, Brush General Requirements
AMS 2750	Pyrometry
AMS 2759/9	Hydrogen Embrittlement Relief (Baking) of Steel Parts
AMSH 6875	Heat Treatment of Steel Raw Materials

(Copies of these documents are available online at www.sae.org.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B499	Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals
ASTM B568	Standard Test Method for Measurement of Coating Thickness by X-Ray Spectrometry
ASTM D3359	Standard Test Methods for Rating Adhesion by Tape Test
ASTM E376	Standard Practice for Measuring Coating Thickness by Magnetic-Field or Eddy-Current (Electromagnetic) Testing Methods
ASTM E466	Standard Practice for Conducting Force Controlled Constant Amplitude Axial Fatigue Tests of Metallic Materials
ASTM F519	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments

(Copies of these documents are available online at www.astm.org.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Application. Alkaline, cyanide free LHE Zn-Ni electroplating meeting this specification is a substitute for cadmium as a corrosion control coating. This process is non-embrittling to high strength steels and can meet the requirements for a non-embrittling process per ASTM F519. LHE Zn-Ni also meets the requirement for lower strength steel alloys, corrosion resistant steel (CRES) alloys, copper-based alloys, and other alloys. The deposit consists of a uniform zinc alloy typically containing 12 to 15% nickel.

3.1.1 Qualification. LHE Zn-Ni electroplating chemical solutions shall be qualified under this specification and shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list in the Qualified Products Database (QPD). The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving qualification. LHE Zn-Ni electroplating processors shall use qualified products located in the QPD. For solution manufacturer qualification, refer to *Section 4: VERFICATION*.

3.1.1.1 Test Methods The monthly and product qualification testing of this specification shall be conducted in accordance with *Table I.*

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TABLE I.
Test Methods

Requirements	**Test Method**	**Testing**	
		Solution Manufacturer Qualification	**Processor Qualification**
Hydrogen Embrittlement	ASTM F519	4.3.1 and 4.3.7.3	3.4.3.3
Corrosion Resistance	ASTM B117	4.3.2 and 4.3.11.3	3.4.3.1
Current Density Versus Composition	XRF Composition Measurements	4.3.3	Not Required
Coating Adhesion	90 Degree Bend Test	4.3.4	3.4.3.2
Paint Adhesion	ASTM D3359 *Method A*	4.3.5	Not Required
Fatigue	AMSH 6875	4.3.6	Not Required
Brush Plating Adhesion	3M #250 Tape Removal	4.3.7.1	Not Required
Brush Plating Corrosion	AMS 2451/9	4.3.7.2	Not Required

3.2 Materials and equipment. Materials and equipment in this section shall be used to accomplish the described process. Reference detailed requirements in the appendix.

3.2.1 Materials. Materials used for plating and conversion coating solutions must be selected from the Qualified Products List QPL-32648.

3.2.2 Equipment.

3.2.2.1 Current. Use either generated or rectified direct current (DC). The ripple value shall ***not*** exceed (five) 5% as measured by dividing the Root Mean Square of the alternating current (AC) voltage component by the DC voltage.

3.2.2.2 Processing tanks. Tanks shall be resistant to the operating temperature and the chemical environment. Tanks in which any electrolytic action is taking place shall be free of electrical shorts and electrically insulated.

3.2.2.3 Transferring solution. If transferring solution outside of plating tank is required for repair or maintenance, all associated equipment used including pumps, hoses, holding tanks, etc., shall be chemically compatible with the solution.

3.2.2.4 Tank temperature. Processing tanks to be operated at temperatures other than room temperature shall be equipped with automatic temperature indicating and regulating devices.

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3.2.2.5 Measuring current. An ammeter shall be placed in series with the LHE Zn-Ni tank cathode. The ammeter shall have sufficient shunts and switches to provide a full-scale reading equal to the maximum capacity of the power source, and an accuracy of ± 5% of the current being measured.

3.2.2.6 Air cleanliness. Air lines shall be suitably trapped and filtered to prevent in-process contamination of the parts to be cleaned.

3.2.2.7 Ovens. Ovens used for hydrogen embrittlement relief must meet AMS 2750.

3.2.2.8 Anodes. Anodes shall be made of Nickel 200 alloy or steel coated with sulfamate nickel. Electroless nickel plated steel shall not be used as anode material. If sulfamate nickel coated steel is used, verify the nickel is non-porous before using. Hot water immersion or ferroxyl testing shall be used to test for nickel porosity.

a. When conformal anodes are used, written instructions shall be developed to instruct technicians on how to use the conformal anode.

b. Hand controlled auxiliary anodes or "wanding" is ***not*** a preferred method. This method shall ***only*** be used when approved by the cognizant engineering authority. A minimum of three (3) plating locations shall be measured after wanding to verify the coating meets the thickness requirements.

3.3 Detailed requirements.

3.3.1 Plating strike. The LHE Zn-Ni shall be deposited directly on the base metal. A strike or pre-plate of an alternate metal is ***not*** allowed except on non-ferrous metals. A short duration high current density strike in the LHE Zn-Ni plating solution is allowed prior to plating at lower current densities.

3.3.2 Prior to plating. Prior to plating LHE Zn-Ni, all machining, forming, welding, and shot peening shall be completed.

3.3.3 Hydrogen embrittlement relief. Parts heat-treated to 180,000 psi ultimate tensile strength and above shall be baked following LHE Zn-Ni plating for a minimum of 23-hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of the initial application of current to the part. Non-standard part material bake times shall be baked in accordance with AMS 2759/9.

3.3.4 Bake interruptions. Bake interruptions are only allowed when the time between the non-compliant temperature and the re-establishment of the specified baking temperature is ***not*** used to determine the total cumulative bake time. A maximum of 30-minutes below specified bake temperature shall be allowed for a bake interruption for each occurrence. Bake interruptions longer than 30-minutes shall be a process failure and the parts shall be subjected to the cognizant engineering authority's review.

3.3.5 Storage of parts. Storage of parts shall be controlled to prevent contact with water or other corrosive materials. Parts shall be stored to permit free air circulation around the parts.

3.3.6 Contact points. Electrical contact between the parts and power source shall be made to prevent chemical or immersion deposition, electrical arcing, and overheating. If parts require plating on the entire surface of the part, contact point locations shall be specified in the technical data. When

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technical data does ***not*** specify contact points, the processor shall develop fixturing and contact point locations to minimize the size and number of bare spots and prevent damage to the part.

NOTE: Contact points with an area greater than 1/16th of a square inch (0.25" by 0.25" square) shall be adjusted during the plating process to eliminate bare spots, or brush plated after the plating process is complete.

3.3.7 Pre-plate stress relief. Steel parts with an ultimate tensile strength of 180,000 psi or greater, shall be stress relieved by baking at 190.5° ±13.9°C (375° ±25°F) for a minimum of four (4) hours. Carburized parts shall be stress relieved by baking at 135° ±13.9°C (275 ±25°F) for a minimum of eight (8) hours. Stress relief is optional for parts with an ultimate tensile strength below 180,000 psi.

NOTE: Stress relief shall precede shot peening, cleaning, and plating.

NOTE: Stress relief is ***not*** required for fasteners, if, all cold working is limited to cold working of the head-to-shank fillet and thread rolling after heat treatment

3.3.8 Processing bath chemistry. Plating and conversion coating solutions shall be maintained according to the manufacturer's technical literature. Deviations from the manufacturer's technical literature are only permitted if the deviations are included in written, locally developed process instructions and requirements in accordance with *Section 3: Requirements.* The processor shall ensure sufficient frequency of chemistry testing to assure the process is within tolerance during processing of parts.

3.3.9 Process changes. No changes shall be made to materials, processes, or controls from the original requirements unless the change is approved by the cognizant engineering authority or the change results in tighter control of the process.

3.3.10 Lot definition. A production lot shall be defined as parts meeting all of the following criteria:

a. Of the same material and part number, or similar part configuration.

b. Plated to the same range of deposit thickness.

c. Plated using the same solutions.

d. Plated in a single continuous plating cycle not to exceed 16-hours.

e. Presented for processor's inspection at one (1) time.

3.4 Quality assurance provisions.

3.4.1 Responsibility for inspection. Unless otherwise specified in this specification the processor shall:

a. Conduct performance of all inspection requirements.

b. Furnish all test samples required for the completion of all inspection and test requirements.

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c. Use contractor or other facility approved by the cognizant engineering authority, to perform the inspection requirements in accordance with this specification.

d. Maintain all documentation relating to inspection requirements for a minimum of seven (7) years.

e. Allow the cognizant engineering authority to perform any inspection in this specification to ensure compliance.

f. ***Not*** submit known defective products when inspection requirements result in successful performance.

g. ***Not*** deviate from any responsibility of contractual compliance for this specification.

3.4.2 Initial processor qualification testing. One full set of monthly qualification tests shall be performed for initial qualification, in addition to the monthly test requirements the processor shall:

a. Perform all tests within 30-days.

b. Ensure all tests pass before production begins.

c. Perform embrittlement test (4.3.1, *TABLE II*) with four (4) specimens instead of two (2).

d. Upon successful completion of the Hydrogen Embrittlement test, one (1) notched tensile specimen shall be sectioned across the notch parallel to the axis of the specimen. Take photomicrographs of the notched area and examine the coating at 80 to 100 times magnification. The coating shall be uniform around the circumference of the notch and the coverage complete. The coating under magnification is rough with voids that may extend to base metal and this is normal and shall not be considered an incomplete plating.

e. Submit a complete analysis report of the plating bath including qualification test results, to the MIL-DTL-32468 Content Manager.

3.4.3 Monthly processor qualification testing. After a tank solution has been qualified, the processor shall:

a. Perform monthly tests with a maximum of 35-days between tests to maintain qualification.

b. Process test specimens using the same procedures and solutions as production parts.

c. ***Not*** re-use test specimens.

d. Retain monthly test data for a minimum of one (1) year.

3.4.3.1 Corrosion resistance.

3.4.3.1.1 Test specimens. Test panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches). A minimum of two (2) test panels shall be used for each test.

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3.4.3.1.2 Specimen preparation. Test panels shall be prepared and processed with *Class 2 Type II* coatings in accordance with *Section 3.7: Zn-Ni application.* Coating thickness shall be determined for each test panel using the same procedure and solution as used on production parts.

3.4.3.1.3 Specimen testing. The test panels shall meet the following criteria:

a. Coatings shall ***not*** show any evidence of corrosion products after 96 hours when tested by continuous exposure to salt spray in accordance with ASTM B117.

b. Areas within 6.35mm (0.25 inches) from the edges of the panel, identification markings, and panel holding points during processing or salt spray exposure shall ***not*** be evaluated.

c. Corrosion testing data shall all be retained for a minimum of one (1) year.

3.4.3.2 Adhesion.

3.4.3.2.1 Test specimens. Test panels shall be new, unused, 4130 steel approximately 1.0 x 25 x 102mm (0.040 x 1 x 4 inches). A minimum of two (2) test panels shall be used for each test.

3.4.3.2.2 Specimen preparation. Test panels shall be prepared and processed with *Class 1 Type II* coatings in accordance with *Section 3.7: Zn-Ni application.*

3.4.3.2.3 Specimen testing. The panels shall be clamped in a vise and repeatedly bent through 0 to 90-degrees until fracture of the base metal occurs then examined at four (4) times magnification for lifting of the deposit along the fractured edge. Crazing of the deposit is acceptable, but flaking-off or lifting of the deposit shall constitute failure.

3.4.3.3 Hydrogen embrittlement.

3.4.3.3.1 Test specimens. Test specimens shall be two (2) round notched 4340 steel specimens in accordance with ASTM F519, Type 1a.1 or 1a.2

3.4.3.3.2 Specimen preparation. Test specimens shall be prepared and processed in accordance with *Section 3.7: Zn-Ni Application*, including embrittlement relief requirements. The coating shall be *Class 1 Type II* plated at a current density of 4.4 to 4.9 A/dm^2 (40 to 45 amperes square feet [ASF]).

3.4.3.3.3 Specimen testing. Verify plating is present in the notch by visual inspection. The plated specimens shall be tested in accordance with ASTM F519. Failure of any one (1) of the specimens shall constitute failure of the test and production shall cease until the cause of the failure is determined and the bath is re-qualified. Acceptance of items completed after the last successfully completed acceptance test shall be withheld until the extent and cause of the failure has been determined.

3.4.4 Finish quality inspection. All parts shall be visually inspected for appearance. If plating parts by lot, measure the thickness of five (5) parts or 10% of the parts, whichever is greater. If plating individual parts, measure the thickness of each part plated. Individual inspection requirements for products produced in lots shall be inspected using a lot sampling plan in accordance with MIL-STD-

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1916, when approved by the cognizant engineering authority. Inspection shall be performed before or after embrittlement relief baking. Inspect the parts in accordance with *Sectioln 3.7 Zn-Ni application*.

3.4.4.1 Appearance. Visual examination of the part shall be conducted after the plating process is complete. The deposit shall be smooth, continuous, adherent to the basis metal, and visually free from porosity, blisters, nodules, pits, and other imperfections detrimental to performance of the plating. Slight staining or discoloration is permissible.

3.4.4.2 Thickness. Non-destructive methods shall be used to determine the thickness on the production parts in three (3) separate areas. All the thickness measurements shall be greater than the minimum specified by the *Class*. Individual measurements greater than 0.0003" over the minimum are acceptable as long as the average of the thickness measurements does not exceed 0.0003" over the minimum.

3.4.4.2.1 Measured area. Thickness shall not be measured within 0.5" of any corners, splines, threads, or other similar feature. On parts that have less than three (3) locations to measure that are not 0.5" away from corners, splines, threads, etc. the processor shall measure the coating thickness. For example, threaded bolts typically do not have any locations that meet the above criteria, therefore the most reasonable option is to measure the head of the bolt, the shaft (if there is an unthreaded portion), and the tip of the bolt (if possible).

3.4.4.2.2 Thickness variation. Plating thickness variations are inherent to the plating process, and recessed areas naturally have thinner Zn-Ni coatings than more exposed areas such as corners. Surfaces that cannot be touched with a 0.75" diameter ball do not need to meet thickness requirements. However, these surfaces shall have a visible Zn-Ni coating. Thickness shall be determined in accordance with ASTM B568, ASTM B499, or equivalent method approved by the cognizant engineering authority, which may include destructive testing.

3.4.4.2.3 Conforming anodes. The use of conforming anodes and or brush plating is allowable to meet the maximum and minimum thickness requirements.

3.4.4.2.4 Bare spots. Brush plating a bare spot to meet maximum and minimum thickness requirements is allowable.

3.4.4.3 Destructive testing. When destructive testing is required, the testing shall be in accordance with the test plan approved by the cognizant engineering authority.

3.4.4.4 Alternative inspection plan. The processor may propose an alternate inspection plan for consideration by the cognizant engineering authority. If approved, this plan shall supersede the preceding inspection plan.

3.4.4.5 Re-qualification. Any significant change in the solution shall require re-qualification of all tests in *Section 3.4.2: Initial processor qualification testing.*

Examples of significant changes are as follows, but not limited to:

a. New solution makeup.

b. Removal of 20% or more of the solution with the accompanying addition of new make-up chemicals.

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c. Storage of the solution for seven (7) days or more outside of the process tank.

d. Shutdown of the process for 30-days or more.

e. Suspected or known contamination and subsequent treatment to remove such contamination.

f. Any other possible events that qualify as significant, inform the cognizant engineering authority.

NOTE: Parts processed in a known or suspected deficient process solution shall ***not*** be presented for acceptance to the cognizant engineering authority.

Examples of ***non***-significant changes:

a. Removal from the process tank to batch "freeze" for carbonate removal - unless left for seven (7) days or more, or when contamination occurs.

b. In-tank dummying to remove metallic contamination.

c. Continual circulation outside of the process tank for carbon treatment, filtration, or carbonate treatment.

3.4.4.6 Acceptance test failure.

3.4.4.6.1 Monthly test failure. If any monthly test results fail to meet the specified requirements, the process is non-conforming.

The processor shall:

a. ***Not*** coat parts until the process is corrected and new specimens are coated and tested with acceptable results.

b. Document all tests results.

c. Inform the cognizant engineering authority.

3.4.4.6.2 Finish quality failure. If results of any finish quality inspection (*Section 3.4.4: Finish quality inspection*) fail to meet specified requirements, the process is non-conforming.

The processor shall:

a. Strip parts with known defects in accordance with *Section 3.5: Stripping*.

b. Re-plate parts in accordance with *Section 3.7: Zn-Ni application*.

c. Inspect other parts in the lot for the non-conforming attributes in accordance with MIL-STD-1916.

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d. Strip and re-process all non-conforming parts as described above. Only passing parts shall be accepted.

3.5 Stripping.

3.5.1 Chemical.

3.5.1.1 Ammonium nitrate. LHE Zn-Ni coating shall be stripped in a solution of ammonium nitrate in accordance with MIL-STD-871. When parts have been stripped and re-plated, the reprocessing shall be documented.

3.5.1.1.1 Supplemental blasting. If black smut remains after stripping, smut shall be removed by light blasting with either plastic media or grit. If garnet or aluminum oxide grit is used, the grit size shall be 100-mesh or finer and the maximum pressure for removal of black smut shall be 40 psi and all surfaces with any coatings shall be masked. If plastic media is used, no masking is required and the grit size shall be 10-grit or finer at a maximum pressure of 60 psi. Pressurized air, hand wiping, or other mechanical means shall be used to remove the smut as long as the method does not damage the component or any coatings on the component such as chromium or nickel.

3.5.1.2 Low strength steel. LHE ZnNi coating on low strength steel parts heat treated less than 180,000 psi ultimate tensile strength, and other materials shall be stripped in accordance with MIL-DTL-871 or MIL-DTL-5002. If MIL-DTL-5002 is followed, the method shall be approved by the cognizant engineering authority, and shall not roughen, pit the basis metal, or adversely affect part dimensions. When parts have been stripped and re-plated, re-processing shall be documented.

3.5.2 Mechanical.

3.5.2.1 Abrasive blasting. Abrasive blasting shall be used in accordance with MIL-STD-1504 using 100-grit media or finer to strip LHE Zn-Ni coating. The method shall remove all the coating and shall not roughen, pit the basis metal, or adversely affect part dimensions. When parts have been stripped and re-plated, re-processing shall be documented.

3.5.2.2 Plastic media blasting. Plastic media blasting shall be used in accordance with MIL-DTL-85891 to strip LHE Zn-Ni coating. The method shall remove all the coating and shall not roughen, pit the basis metal, or adversely affect part dimensions. When parts have been stripped and re-plated, re-processing shall be documented.

3.6 Cleaning and masking.

The processor shall:

a. Clean and degrease all parts as necessary prior to abrasive blast or plating.

b. Mask parts to protect surfaces where coating is not required or other areas where blasting is prohibited by technical data. All surfaces that have an existing coating (such as chrome, nickel, or tungsten carbide cobalt) shall be masked prior to blasting.

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c. Abrasive blast surfaces to be LHE Zn-Ni plated using 100-grit media or finer in accordance with MIL-STD-1504.

d. Minimize contamination after blasting by limited handling and contact with the blasted surfaces. When handling is required, use clean gloves and clean surfaces when transporting parts.

e. Mask parts as needed to prevent plating where technical data prohibits plating. The same masking used for blasting operations may be used for plating.

3.7 Zn-Ni application.

The processor shall:

a. Start plating process within four (4) hours of the abrasive blast.

b. When time between abrasive blast and plate exceeds four (4) hours, re-blast parts prior to plating.

NOTE: Other mechanical and chemical activation processes (such as alkaline cleaning, rinses, or dilute acid activation) are acceptable to replace abrasive blast or be used in addition to abrasive blast if all the requirements of *Section 3: Requirements* are met and the alternate methods are approved by the cognizant engineering authority.

c. Immerse parts into the plating solution.

NOTE: An optional strike at 60 to 90 ASF for up to 60-seconds may be applied before reducing the current to the target current density.

d. Plate at a current density of 30 to 60 ASF (45 ASF optimum) for sufficient time (typically 15-minutes for *Class 2*) to form a deposit that meets the minimum thickness required for the part.

e. Turn off the current and remove the part from the solution - as soon as possible - as the plating solution slowly dissolves the applied coating when no current is applied.

f. Rinse the parts in clean water.

g. Inspect the part's surface for continuous plating. If bare spots are found, adjust the fixturing and or orientation and return the part to the plating tank and repeat the process in accordance with the procedures in this section.

NOTE: If the Zn-Ni coating dries during the process without conversion coat, it passivates. Processor should keep coating wet for best results. At the discretion of the processor, the inspection step may be performed after the conversion coating

h. Conversion coating.

1 - If *Type I* is specified, dry with clean compressed air and do ***not*** conduct conversion coating step.

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2 - If *Type II* is specified, immerse the part in the conversion coating solution in accordance with the solution manufacturer's technical literature. Rinse the part in clean water and dry with clean compressed air.

NOTE: Embrittlement relieve high strength steel parts (>180 ksi) at 190.5° ±13.9°C (375° ±25°F) for a minimum of 23-hours within four (4) hours of the initial application of current to the part. It is acceptable to bake low strength steel parts, but not required.

3.8 Re-processing. If the applied coating is defective and brush plating cannot correct the deficiency, or if brush plating is not economical, the coating shall be stripped and reapplied in accordance with the procedures in *Section 3.7: Zn-Ni application*.

3.9 Brush plating.

The processor shall:

a. Process parts or areas requiring brush plating in accordance with AMS 2451/9 or MIL-STD-865.

b. Use non-embrittling Zn-Ni products on the QPL and approved by the cognizant engineering authority.

c. ***Not*** use pre-plating strike (Nickel Strike).

d. Brush plate directly onto the bare metal after surface preparation.

4. VERIFICATION

4.1 Plating solution qualification. This section covers acceptance criteria for plating solutions used for the deposition of a Zn-Ni coating and subsequent treatment of the coating on ultra-high strength steel (180 ksi) ultimate strength and above, utilizing a LHE Zn-Ni electroplating process with a trivalent chrome or chromium free conversion coat. The plating solution manufacturer shall:

a. Meet the requirements of this *Verification* section.

b. Test each batch of plating solution in accordance with this *Verification* section.

c. Provide a Certificate of Conformance to the users of the solution.

d. Maintain test records available upon request for a minimum of 10-years.

e. Submit test data to qualify activity in accordance with *Section 6.3: Qualification.*

4.2 Procedures.

4.2.1 Testing required. This section outlines the testing required to qualify LHE Zn-Ni solutions for use in accordance with this specification. If the formulation changes, manufacturing equipment changes, technical literature changes, or manufacturing location changes the initial qualification testing must be re-completed and accepted by the cognizant engineering authority using the new conditions before solution is supplied to customers. Minor changes to manufacturing equipment or

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technical literature will be approved in writing by the MIL-DTL-32648 Content Manager without full re-qualification.

4.2.2 Tested as a pair. The Zn-Ni plating solution and the conversion coating solution shall be tested as a pair. The plating solutions and conversion coating solutions shall not be qualified independently.

4.2.3 Using same procedures. The test specimens for all testing shall be prepared and plated using the same procedures specified within the manufacturer's technical literature. If activation or strike steps are required in the manufacturer's technical literature, testing shall be accomplished with the activation or strike step(s) included. If the activation or strike steps are defined as "optional" within the manufacturer's technical literature, the testing shall be accomplished without the optional step(s). Unless otherwise specified, where the technical literature provides a range for a processing parameter (such as current density or plating bath temperature) the middle of the range shall be used for qualification testing.

4.2.4 Coating thickness and composition. Coating thickness and composition shall be determined in accordance with ASTM 7091. During plating, the specimens shall be mounted on a rack to ensure uniform plating. All areas of the rack, except the contact area, shall be coated with a suitable masking. An ammeter having a sensitivity of 0.5 amperes or better shall be connected between the specimen rack and the cathode.

4.2.5 Directly to base metal. The LHE Zn-Ni coating shall be applied directly to the base metal without the use of another coating between the base metal and the LHE Zn-Ni coating.

4.2.6 Finish quality. All test specimens used for qualification shall be visually inspected after plating. The deposit shall be smooth, continuous, adherent to the basis metal, and visually free from porosity, blisters, nodules, pits, and other imperfections detrimental to performance of the plating. Slight staining or discoloration is acceptable.

4.3 Qualification testing for plating and conversion coating solutions.

4.3.1 Hydrogen embrittlement.

4.3.1.1 Test specimens. Test specimens shall be manufactured in accordance with ASTM F519, Type 1a.1 or 1a.2.

4.3.1.2 Specimen preparation. Specimens shall be plated using low, middle, and high current densities and compositions in accordance with *TABLE II* to test across the range of bath compositions and current densities and follow defined current density and composition ranges in accordance with the manufacturer's technical literature. The specimens shall be plated to a thickness of 0.015 to 0.020mm (0.0006 to 0.0008 inch) and shall be *Type II* conversion coated. The specimens shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of the initiation of current application. Four (4) specimens shall be prepared in accordance with the test condition *TABLE II*.

4.3.1.3 Specimen testing.

a. Test specimens shall be subjected to a sustained load test at 75% of the ultimate notched tensile strength in accordance with ASTM F519 for a minimum of 200-hours.

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b. The test shall be considered passed if all specimens from each composition group do not break or show any cracking.

c. The test shall be considered a failure if any one (1) of the specimens breaks or cracks during testing. One (1) repeat of four (4) additional specimens shall be carried out for each condition that failed the original testing.

d. Upon successful completion of the sustained load test, at least one (1) of the notched tensile specimens from each current density/composition set shall be sectioned across the notch parallel to the axis of the specimen. Photomicrographs shall be taken of the notched area and the coating examined at 80 to 100 times magnification. The coating shall be uniform around the circumference of the notch and the coverage complete. The coating under magnification is rough with voids that may extend to the base metal; this is normal and shall not be considered an incomplete plating.

TABLE II.
Composition and current density ranges.

Compositions	Current Density Ranges			Total Number of Coupons
	Low	Middle	High	
High Zn/High Ni	4	4	4	12
High Zn/Low Ni	4	4	4	12
Low Zn/High Ni	4	4	4	12
Low Zn/Low Ni	4	4	4	12
Med Zn/Med Ni	4	4	4	12

4.3.2 Corrosion resistance.

4.3.2.1 Test specimens. Panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches).

4.3.2.2 Specimen preparation. The test panels shall be plated to 0.0051 to 0.0102mm (0.0002 to 0.0004 inches) coating thickness. Test panels for *Type II* and Painted Corrosion testing shall be conversion coated. Plating and conversion coating shall be in accordance with the manufacturer's technical literature.

Type I. Four (4) test panels shall be plated at the lower, middle, and upper limits of the manufacturer's technical literature current density range. These panels may double as the test panels for the current density versus composition test in *Section 4.3.3: Current density versus composition*.

Type II and Painted Corrosion. Twelve (12) test panels shall be plated at the middle of the manufacturer's technical literature current density range and shall then be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F).

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a. Painted Corrosion panels testing surface shall be painted in accordance with MIL-PRF-85582 or MIL-PRF-23377 primers, *Class C* and *Class N* coatings.

b. Six (6) panels shall be coated with *Class C* primer.

c. Six (6) panels shall be coated with *Class N* primer, to a thickness in accordance with the primer manufacturer's technical literature.

4.3.2.3 Specimen testing.

4.3.2.3.1 Type I. Coatings shall show no evidence of red rust corrosion products when exposed to salt spray for 1,000 hours in accordance with ASTM B117.

4.3.2.3.2 Type II. Coatings shall show no evidence of white corrosion products after 96 hours and no evidence of red corrosion on the basis metal after 1,000 hours when tested by continuous exposure to salt spray in accordance with ASTM B117.

4.3.2.3.3 Painted corrosion. The test panels shall be scribed in accordance with MIL-PRF-32239.

4.3.2.3.3.1 Class C primers. Test panels shall show no evidence of red corrosion on the basis metal after 2,000 hours when tested by continuous exposure to salt spray in accordance with ASTM B117.

4.3.2.3.3.2 Class N primers. Test panels shall show no evidence of primer blistering after 2,000 hours when tested by continuous exposure to salt spray in accordance with ASTM B117. Corrosion in the scribe is acceptable for *Class N* primers as long as the corrosion does not cause the primer to blister.

4.3.2.3.4 Additional acceptance criteria. Areas within 0.25 inch (6.35mm) from the edges of the panel, the identification markings, and the panel holding points during processing or salt spray exposure, shall not be evaluated and corrosion in those areas shall not be cause for rejection. All corrosion tests shall be carried out using a 5% salt solution. Differences in color between the test panels shall not be cause for rejection. The test shall be considered a failure if any one (1) of the specimens fails during testing.

TABLE III.
Schedule for corrosion resistance properties.

Panels at high current density	Panels at middle current density	Panels at low current density	Thickness	Type	Material to be coated	Salt spray exposure (hours-White Corrosion)	Salt spray exposure (hours-Red Rust)
4	4	4	0.0002" to 0.0004"	I	4130 Steel	-	1000
-	12	-	0.0002" to 0.0004"	II	4130 Steel	96	1000
-	12	-	0.0002" to 0.0004"	Painted 6 Class C 6 Class N	4130 Steel	-	2000

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4.3.3 Current density versus composition.

4.3.3.1 Test specimens. Corrosion testing panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches).

4.3.3.2 Specimen preparation. Specimens shall be plated using low, middle, and high current densities and compositions in accordance with *TABLE IV* to test across the range of bath compositions and current densities. The current density and composition ranges shall be in accordance with manufacturer's technical literature. The specimens shall be plated with a *Class 2 Type I* coating and shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of the initiation of current application. Two (2) specimens shall be prepared in accordance with test condition *TABLE IV*.

TABLE IV.
Composition and current density ranges.

Compositions	Current Density Ranges			Total Number of Coupons
	Low	Middle	High	
High Zn/High Ni	2	2	2	6
High Zn/Low Ni	2	2	2	6
Low Zn/High Ni	2	2	2	6
Low Zn/Low Ni	2	2	2	6
Med Zn/Med Ni	2	2	2	6

4.3.3.3 Specimen testing. Eight (8) points of each specimen shall be tested in the approximate locations shown in *FIGURE 1*. Composition shall be determined by X-ray Fluorescence (XRF) or other method approved by the cognizant engineering authority, and shall be within 12 to 15% nickel composition range.

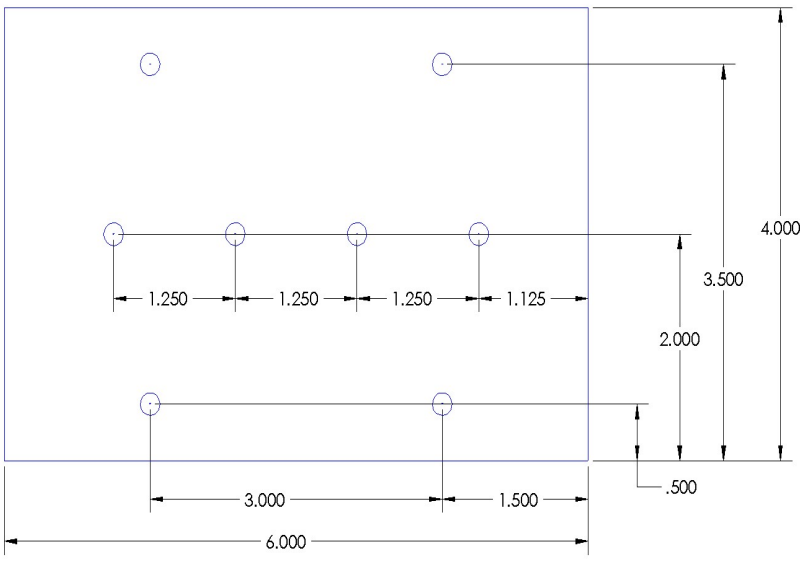


FIGURE 1. Recommended composition test points.

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4.3.4 Coating adhesion.

4.3.4.1 Test specimens. Adhesion panels shall be new, unused, 4130 steel approximately 1.0 x 25 x 102mm (0.040 x 1 x 4 inches).

4.3.4.2 Specimen preparation. Specimens shall be LHE Zn-Ni plated in accordance with the manufacturer's technical literature. Eight (8) panels shall be plated with a *Class 3 Type I* coating and baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F).

4.3.4.3 Specimen testing. Test specimens shall be clamped in a vise and bent repeatedly to approximately 90 degrees and back to the original position until rupture of the basis metal occurs. The fractured surface shall be examined unaided or at up to 10X magnification for lifting of the deposit along the broken edge. Lifting of the deposit on any one (1) specimen shall constitute failure.

4.3.5 Paint adhesion.

4.3.5.1 Test specimens. Adhesion specimen panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches).

4.3.5.2 Specimen preparation. Specimens shall be LHE Zn-Ni plated and conversion coated in accordance with the manufacturer's technical literature and baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F).

a. Each panel shall be plated with *Class 1 Type II* coatings.

b. Four (4) test panels shall be prepared only with primer in accordance with MIL-PRF-85582.

c. Four (4) test panels shall be prepared with primer and topcoat in accordance with MIL-PRF-85582 or MIL-PRF-23377 for primer, and MIL-PRF-85285 for topcoat.

4.3.5.3 Specimen testing.

a. Test panels shall be immersed in distilled water for 24-hours at room temperature.

b. Remove test panels from water and wipe or pat dry with a soft cloth.

c. Within three (3) minutes after removal from water, make two (2) parallel scribes with a stylus through the coating to the substrate approximately ¾ of an inch apart and approximately two (2) inches long.

d. The panels shall be scribed to the substrate from opposing ends of the parallel scribes to form an "X". Bare steel must be visible in the bottom of the entire scribe. If the first scribe did not penetrate to the base metal on the entire scribe, it is acceptable to re-scribe along the original scribe or in a new un-scribed area.

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e. Immediately apply a one (1) inch wide strip of masking tape with the adhesive side down across the scribes. Press the tape against the surface of the coating by passing a 4 and a ½ pound rubber covered roller, approximately 3 and ½ inches in diameter and 1 and ¾ inches in width across the tape eight times. Remove the tape with single quick motion and examine for coating damage. The masking tape shall have an average adhesion of at least 60 ounces/inch width (3M Company #250 or equivalent as approved by the cognizant engineering authority for the paint adhesion.

f. There shall be no peeling or delamination between the primer and topcoat or at the primer and substrate interface. Some trace peeling is acceptable along the scribe lines. However, any coating removal more than 1/32 of an inch shall be cause for rejection. Poor adhesion on any one (1) specimen shall constitute failure.

g. The scribing, tape test, and evaluation shall be performed in accordance with ASTM D3359 *Method A*, in lieu of the procedures above. If ASTM D3359 *Method A* is used, the panels shall achieve a 4A rating or better.

TABLE V
Schedule for paint adhesion properties.

Minimum number of panels	Class	Type	Material to be coated	Paint Application
4	1	II	4130 Steel	Primer Only
4	1	II	4130 Steel	Primer and Topcoat

4.3.6 Fatigue

4.3.6.1 Test specimens.

a. Specimens shall be manufactured in accordance with *FIGURE 2*.

b. Coupons shall be manufactured from 300M heat-treated to 1930 to 2068 megapascal (MPa) (280 to 300 ksi) in accordance with AMSH 6875 and tested at the stress loads in *TABLE VI*.

c. Fatigue data will be reviewed by the MIL-DTL-32648 Content Manager.

d. Fatigue data results shall not have statistical differences in fatigue performance compared to baseline cadmium plated fatigue coupons.

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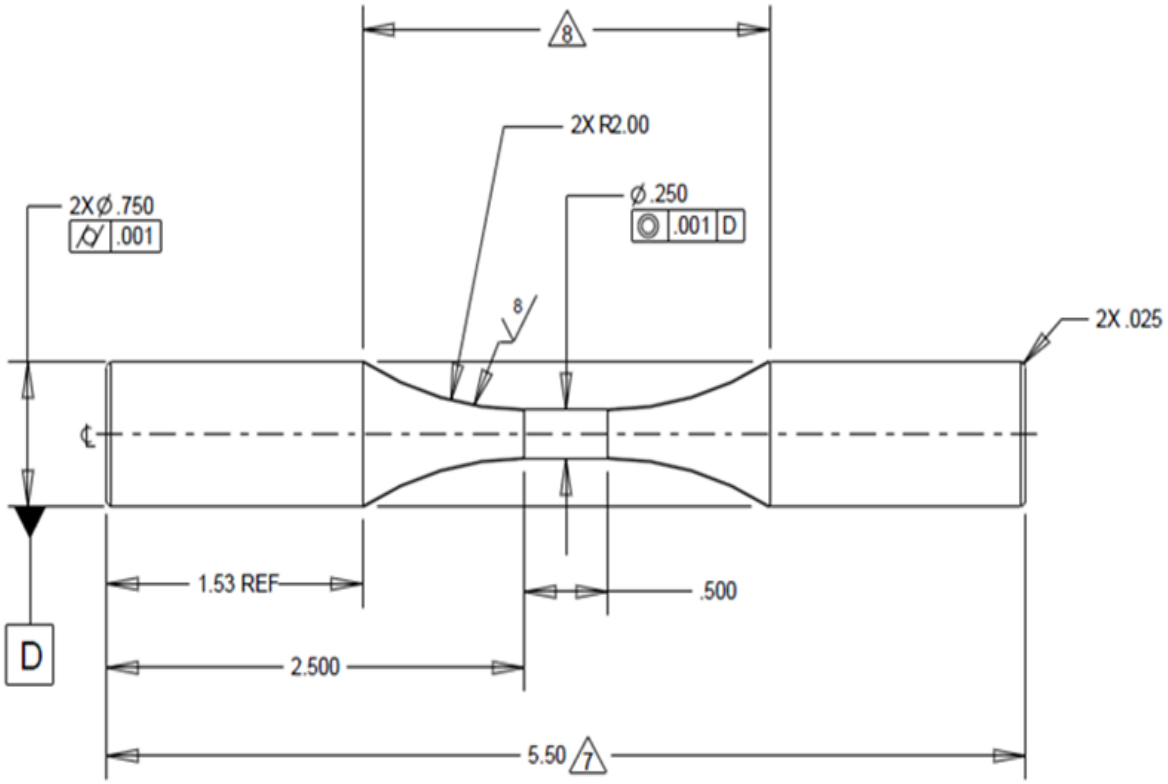


FIGURE 2. Fatigue test coupon.

NOTES:

7 Shot peen in accordance with SAE AMS-2430, 8-10A, 100% coverage, and S230 shot size and all shot peening shall be computer controlled

8 Do ***not*** shot peen area when preparing un-peened coupons for testing.

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4.3.6.2 Specimen preparation.

a. No plating. 54 specimens shall be left bare.

b. Cadmium. 54 specimens shall be cadmium plated in accordance with MIL-STD-870 *Class 1 Type II* or QQ-P-416 *Class 1 Type II*.

c. LHE Zn-Ni. 54 specimens shall be LHE Zn-Ni plated and conversion coated in accordance with the manufacturer's technical literature.

NOTE: During plating, both grip ends shall be masked such that in only the gage section of the coupon that receives the coating.

d. Specimens shall be plated to a thickness of 0.015 to 0.020mm (0.0006 to 0.0008 inch).

e. Specimens shall ***not*** be baked for more than 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of removal from the bath.

4.3.6.3 Specimen testing.

a. Fatigue coupons shall be tested in accordance with ASTM E466 and in accordance with the stress loads in *TABLE VI*.

b. Fatigue data will be reviewed by the cognizant engineering authority.

c. Fatigue data shall result in no statistical difference in fatigue performance using Weibull distribution as compared to baseline cadmium plated fatigue coupons, or the uncoated specimens.

TABLE VI
Fatigue test matrix.

	Stress Loads (ksi) R= 0.1			Stress Loads (ksi) R= -1			Total Quantity
	150	180	220	150	180	220	
Bare	9	9	9	9	9	9	54
LHE Zn-Ni Plated	9	9	9	9	9	9	54
LHE Cd Baseline	9	9	9	9	9	9	54
Total Fatigue Coupons							162

4.3.7 Brush plating. Brush plating solutions shall meet the requirements of this section. Immersion solutions are ***not*** required to meet the requirements of this section.

4.3.7.1 Adhesion testing.

4.3.7.1.1 Test specimens. Testing shall be performed on three (3) new, unused, 4130 steel approximately 1.0 X 25 X 153mm (0.040 X 4 X 6 inches).

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4.3.7.1.2 Specimen preparation.

a. Coupons shall be prepared by brush plating over the following simulated damaged substrates:

1. Class 1 Type II. LHE Zn-Ni in accordance with manufacturer's technical literature.

2. Class 1 Type II. Cadmium in accordance with MIL-STD-870 or AMS-QQ-P-416.

NOTE: All of the qualified tank solutions shall be tested. For example, if there are three (3) solution/ conversion coating pairs qualified, all three (3) solution pairs shall be tested for a total of four (4) tests, three (3) for Zn-Ni and one (1) for cadmium.

b. Simulated damage area shall be created by masking off a ½ inch strip lengthwise down the center of the coupon prior to plating using the standard plating tank.

c. Masking shall be removed after plating exposing a bare strip to be brush plated.

d. Specimens shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of removal from the bath prior to brush plating.

e. Masked surface shall be cleaned to remove any adhesive tape residue.

4.3.7.1.3 Specimen testing. Apply a one (1) inch wide strip of 3M Company #250 Masking Tape, or equivalent, to the brush plated surface along the length of the coupon. Apply tape firmly, making sure all edges adhere. Remove the tape with one quick vertical motion. Any plating adhering to the tape shall be rejected as failure.

4.3.7.2 Corrosion testing.

4.3.7.2.1 Test specimens. Testing shall be performed on four (4) new, unused panels of 4130 steel approximately 1.0 x 25 x 102mm (0.040 x 1 x 4 inches).

4.3.7.2.2 Specimen preparation. Specimens shall be prepared in accordance with AMS 2451/9.

4.3.7.2.3 Specimen testing. Specimens shall be tested in accordance with AMS 2451/9.

4.3.7.3 Hydrogen Embrittlement testing.

4.3.7.3.1 Test specimens. Testing shall be performed on four (4) specimens in accordance with ASTM F519, Type 1a.1 or 1a.2.

4.3.7.3.2 Specimen preparation. Specimens shall be prepared in accordance with AMS 2451/9.

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4.3.8 Control factors. The solution manufacturer for immersion plating solution or brush plating solution shall include the following processing and control information within the product's technical literature:

a. Surface preparation including strike, if used (this strike refers to a high current density plating strike in the same LHE Zn-Ni solution used for plating).

b. Detailed processing procedures.

c. Plating bath composition and composition control limit.

d. Analytical procedures.

e. Impurity limits.

f. Impurity treatment procedures.

g. Impurity removal treatments.

4.3.9 Instructions for qualification testing submission.

4.3.9.1 Analysis report. A complete analysis report shall be submitted in digital hardcopy (DVD or CD) and via encrypted email to the MIL-DTL-32648 Content Manager.
The analysis report shall include:

a. Plating bath.

b. Qualification test results.

c. Product technical literature.

4.3.10 Plating solution qualification test summary.

4.3.10.1 Required testing. Listed by test and requirement paragraph:

a. Hydrogen Embrittlement 4.3.1

b. Type I, Type II, and Painted Corrosion 4.3.2

c. Current Density versus Composition 4.3.3

d. Coating Adhesion 4.3.4

e. Paint Adhesion 4.3.5

f. Fatigue 4.3.6

g. Brush Plating 4.3.7

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4.3.11 On-going qualification testing.

4.3.11.1 Internal quality systems. The solution manufacturer shall have internal quality systems in place to ensure the solutions have consistent performance.

4.3.11.1.1 Regular testing. The solution manufacturer shall perform testing in accordance with this section every five (5) years and submit the results to the cognizant engineering authority in order to maintain qualification. It is acceptable for the processor to submit monthly qualification testing (3.4.3).

4.3.11.2 Hydrogen Embrittlement.

4.3.11.2.1 Test specimens. Test specimens shall be manufactured in accordance with ASTM F519, Type 1a.1 or 1a.2.

4.3.11.2.2 Specimen preparation. Four (4) specimens shall be plated to a thickness of 0.015 to 0.020mm (0.0006 to 0.0008 inch) and shall be *Type II* conversion coated. The specimens shall be plated using the lower limit of the current density range. The specimens shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F) within four (4) hours of the initiation of current application.

4.3.11.2.3 Specimen testing. Test specimens shall be subjected to a sustained load test at 75% of the ultimate notched tensile strength in accordance with ASTM F519 for a minimum of 200-hours. The test shall be acceptable when all specimens from each composition group do not break or show any cracking. The test shall be rejected as failure when any one (1) of the specimens breaks or cracks during testing.

4.3.11.3 Corrosion.

4.3.11.3.1 Test specimens. The corrosion test panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches).

4.3.11.3.2 Specimen preparation. The test panels shall be plated to 0.0051 to 0.0102mm (0.0002 to 0.0004 inches) coating thickness and shall ***not*** be conversion coated *Type I*. Two (2) test panels shall be plated at the lower, middle, and upper limits of the manufacturer's technical literature current density range. These panels may double as the test panels for the current density versus composition test. The test panels shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F).

4.3.11.3.3 Specimen testing. Coatings shall not show evidence of red rust corrosion products when exposed to salt spray for 1,000 hours in accordance with ASTM B117.

4.3.11.4 Composition.

4.3.11.4.1 Test specimens. Composition test panels shall be new, unused, 4130 steel approximately 1.0 x 102 x 153mm (0.040 x 4 x 6 inches).

4.3.11.4.2 Specimen preparation. The test panels shall be plated to 0.0051 to 0.0102mm (0.0002 to 0.0004 inches) coating thickness and shall ***not*** be conversion coated *Type I*. Two (2) test panels shall be plated at the lower, middle, and upper limits of the manufacturer's technical literature current density range. These panels may double as the test panels for the

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corrosion test. The test panels shall be baked for 23 ±1 hours at 190.5° ±13.9°C (375° ±25°F).

4.3.11.4.3 Specimen Testing. Eight (8) points of each specimen shall be tested in accordance with the locations shown in *FIGURE A-1*. Composition shall be determined by X-ray Fluorescence (XRF) or by another method approved by the cognizant engineering authority and shall be within 12 to 15% nickel composition range.

4.3.12 Retention of qualification. To retain qualification approval of products listed on the Qualified Products List (QPL) in the Qualified Products Database (QPD), the manufacturer will be required to verify by certification to the qualifying activity that its product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in five-year intervals from the date of original qualification. The certification action will be initiated by the qualifying activity.

4.3.12.1.1 Requirements. To retain qualification of a product approved to include in this specification, the plating solution manufacturer shall:

a. Submit a certification to the cognizant engineering authority with verification the processor's product complies with the requirements of this specification.

b. Ensure no significant changes have been made in, formulation, raw materials or supplier(s) of raw materials, methods of manufacture, equipment, or geographic location, without prior written cognizant engineering authority approval. A significant change, when determined by the cognizant engineering authority, is a change that could affect the properties or performance of the parts.

c. Allow the cognizant engineering authority, at any time, re-examination of the qualified product to determine the product continues to meet any or all of this specification's requirements.

4.3.12.2 Information required. The cognizant engineering authority will request the following when authorizing the forwarding of qualification samples.

The plating solution manufacturer shall submit:

a. Qualification inspection sample.

b. Test report(s) showing the materials are in accordance with the requirements of this specification.

c. Safety Data Sheet(s) (SDS) prepared and submitted in accordance with FED-STD-313 and 29 CFR 1910.1200. The 29 CFR 1910.1200 requires a SDS for each hazardous chemical used in an operation, and each SDS must be readily available to personnel using the material. All questions pertinent to the effect(s) of these plating and conversion coatings on the health of personnel using them shall be addressed in the SDS(s). The Government will refer to the SDS(s) when contacting the appropriate medical services who will act as the Government's adviser. Contracting officer will identify the activities who require copies of SDS(s).

4.3.12.3 Conformity to qualified sample. All materials supplied under this specification shall be manufactured using the same formulation, raw materials and supplier(s) of raw materials,

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methods of manufacture, equipment, and geographic location the same as the qualification sample. Unless changes have been approved by the cognizant engineering authority.

5. PACKAGING

5.1 Plated parts. Shall be handled and packaged to ensure the physical characteristics and properties of the plate are preserved.

5.2 Shipping. Packages of plated parts shall be prepared for shipment in accordance with commercial practice and in accordance with applicable rules and regulations pertaining to the handling, packaging, and transportation of parts to ensure carrier acceptance and safe delivery.

5.3 Acknowledgment. The processor shall include identification of this specification and any revision(s) in all quotes and when acknowledging purchase orders.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful but is not mandatory.

6.1 Intended use. This MIL-DTL was developed as an alternative process to use in place of low embrittling cadmium plating. It covers the processes and materials required for the electro-deposition of Low Hydrogen Embrittling (LHE) Zinc-Nickel (Zn-Ni) alloy on high strength steel substrates and heat treating techniques needed to ensure low embrittlement of steel.

6.2. Acquisition requirements. Specify the following in the acquisition document:

a. Plating *Type* and *Class*.

b. Method for determining plating thickness.

c. Quality sampling plan if different from the requirement in *Section 3.4: Quality assurance provisions*.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion on the Qualified Products List QPL-32648 in the Qualified Products Database. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the AF-70 Hill AF Base, AFMC 417 SCMS/GUEA Landing Gear, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense. Refer to the SD-6 for product qualification procedures. And, visit the Qualified Products Database at https://qpd.dla.mil/qpd/.

6.3.1 Processor requirements. Processors are not required to follow the requirements in *Section 4: Verification*. However, processors are to use LHE Zn-Ni plating solutions and conversion coating solutions listed on the Qualified Products List QPL-32648.

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6.4 Subject term (key word) listing.

Corrosion resistance
Electrodeposited
High strength steel
Low strength steel
Content manager

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 70

Preparing Activity
Air Force – 70
(Project MFFP-2019-002)

Review Activities:
Army – AV
Navy – SH
Air Force – 20
DLA – CC

NOTE: The activities listed above were interested in this specification as of the date of this specification. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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